

Dark Matter Fraction Analysis Project Tutorial

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Introduction

This tutorial will step you through the process of deriving the dark matter fraction and plotting it against various galaxy parameters. This tutorial is using the *Spitzer Photometry and Accurate Rotation Curves* (SPARC) dataset provided by [Lelli et al. 2016](#). This dataset comprises 175 disk galaxies and has two tables of info for each galaxy. Table 1 includes all the galaxy properties that will be used to compare with the f_{DM} . Table 2 includes variables that will be used to derive the f_{DM} , notably the observed rotation velocity (V_{obs}), rotation velocity of gas (V_{gas}), rotation velocity of the stellar disk (V_{disk}), and rotation velocity of the stellar bulge (V_{bulge}).

The first part in the project is deriving the dark matter fraction inside of TOPCAT. Start by downloading the file included in the github ([or here](#)). Open TOPCAT and navigate to “Load Table” in the top left. After importing the tables, the data can be viewed and plotted. You can plot by clicking the “Plane Plot” button. For rotation curves, use radius and observed velocity. You can add the Keplerian Decline as a function by adding $\sqrt{G_kpc * M_star / x}$. This will give you a good idea of what the contents of dark matter is in a galaxy depending on the gap between the predicted and observed.

Actually finding the dark matter fraction is what is most important in this analysis. Start by adding columns for Table 2 by going to “Column Info.” Here you will add a new column named V_{bar} , which is the baryonic mass rotation velocity. The expression is “ $\sqrt{V_{gas} * V_{gas} + V_{disk} * V_{disk} + V_{bulge} * V_{bulge}}$ ”. Next is V_{dm} , which is the velocity of dark matter. The expression is “ $\sqrt{abs(V_{obs} * V_{obs} - V_{bar} * V_{bar})}$ ”. Finally, there is the dark matter fraction and the expression is “ $(V_{dm} * V_{dm}) / (V_{obs} * V_{obs})$ ”. For better visual, the equations are as follows:

$$V_{DM}^2 = V_{obs}^2 - (V_{gas}^2 + V_{disk}^2 + V_{bulge}^2) \quad f_{DM} = \frac{V_{DM}^2}{V_{obs}^2}$$

Now that all the dark matter fraction has been derived, you can start the hardest part of this analysis. To be able to compare the dark matter fraction with the values of the other table, the fraction needs to be in its own column in Table 1. This must be done manually and takes the most time for this tutorial. Start by adding a column in Table 1 called F_{DM} (or whatever name you prefer), then type in the expression “ $NULL * 0.0$ ”. This creates an empty column as a double. Then go through each galaxy in Table 2 at the largest radius observed and find the f_{dm} that was

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calculated. Then type that value into the corresponding galaxy's FDM. This is very tedious and I was not able to find a better solution, so doing it this way ensures you can plot the data correctly and easily.

You can now start plotting your results! Table 1 includes luminosity ($3.6\ \mu\text{m}$), effective radius, effective surface brightness, scale stellar disk length, effective surface brightness on the disk, total HI mass, radius of HI, and flat rotation velocity. It is recommended to put this data in log scale by creating a new column and use the expression " $\log_{10}(\text{value})$ ". When you make a plot, use the dark matter fraction as the Y axis and the wanted variable for X. To add a trendline, go to "Form," then add a "Linear LinearFit." The correlation, slope, intercept, and deviation are all in the linear fit section for the plot.

That is all for this tutorial, everything shown is what was done for data analysis in my Dark Matter in Galaxies paper. There is a separate repository for my data results if you want to check or compare with what you were able to do with this tutorial.